

Using an appropriate range of Expression

This is an extremely important aspect for getting the optimal expressiveness and realism with virtual instruments.

All real physical instruments require a much greater degree of effort to achieve the highest possible dynamics for that instrument. This is not the case for most physical MIDI controllers; it is indeed very easy to push an expression pedal, or a slider, to the maximum and leave it there forever. Oddly enough, even breath and wind controllers tend to jump very easily to the highest dynamics (unless carefully set), so that the most expressive range (mp to mf) is seldom exploited. This results all too often in the virtual instrument being played in an excessive range of dynamics, leading unavoidably to a poor, unrealistic performance.

Please keep in mind that the key to expressiveness and realism is a proper use of dynamics. Always monitor the input expression value on the main display and adjust the dynamics accordingly. A useful trick is to set the overall monitoring volume fairly high. This automatically prevents any inclination toward over using the higher levels of dynamics.

The Expression slider on the main GUI becomes “hot”, i.e. with a red halo surrounding it, if the level of expression stays constantly above 75%. When you see such a red alert warning, please consider avoiding such extensive use of higher dynamics.



It's also important to optimize the general volume of your loudspeakers/headphones at an average value of Expression. Too low an output level would easily induce the user to use high Expression levels, usually yielding excessively “biting” sounds.

Introduction

SWAM Solo Strings is a collection of solo string instruments conceived and developed by Stefano Lucato and the Audio Modeling team. These instruments are based on Physical Modeling technology and do not make use of any samples.

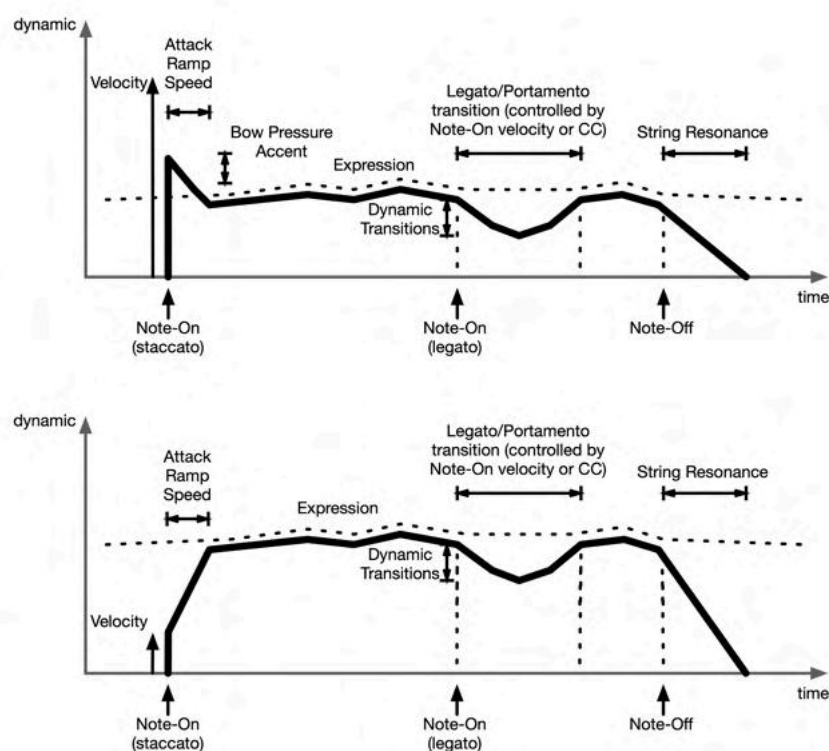
The SWAM Solo Strings collection includes four string instruments: Double Bass, Cello, Viola and Violin.

SWAM Solo Strings instruments come as a Standalone application, VST, VST3, Audio Units and AAX plug-ins for Desktop, and Standalone App, AUV3 and IAA plug-ins for iOS.

Any SWAM Solo Strings product can be virtually played using a bow or pizzicato with a finger.

In addition to standard SWAM controls such as staccato / legato (with velocity- or CC-controlled portamento time), dynamic expression and vibrato depth and frequency, there are several additional parameters to control the instrument behavior which are specific to bowed or pizzicato string instruments.

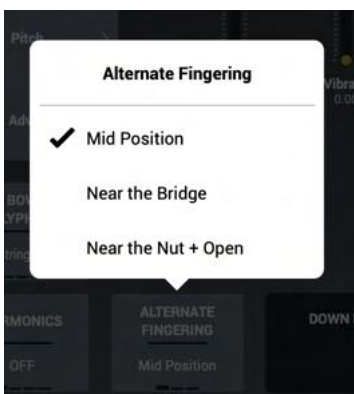
NOTE: when attack is controlled by Note On velocity, the Velocity value acts mainly on Bow Pressure, i.e. dynamic attack is not proportional to the Velocity value, as for the Woodwinds and Brass.



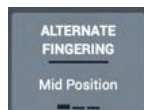
Interaction with the product

You can control the parameters of SWAM instruments directly through the user interface (UI) using a mouse, finger on touch devices, or other pointing systems like Eye Trackers or OCR. Additionally, interaction is possible via Screen Reader, [keyboard shortcuts](#), or [MIDI](#) for parameters that are mappable and effectively mapped. All SWAM products have been validated by visually impaired users for a high level of accessibility.

Parameters are graphically represented by knobs, sliders, button rows with all options explicitly listed, or toggle buttons. For toggle buttons, you can press and hold to get an explicit list of values.



These buttons also have a small graphical indication of the number of available options via hatching, with the selected value highlighted through a more pronounced mark.



The interface is designed for various uses on different screen sizes, devices, and operating systems. The [Main "... menu](#) on the home screen acts as a guided wizard that provides access to almost all functions.

Next to each object, the contextual "... menu offers access to the most important and contextual options, encouraging users to explore further.

System parameters are categorized on the home screen into different groups to make it easier for less experienced users or infrequent users to understand and get started. However, the experienced and frequent user has access to the [Search](#) button located at the top of the header. This tool provides at the same time an overview of all parameters and a search function to quickly find what is not immediately visible.

Main window

The main window shows the most important parameters, instrument monitors and animations.



Instrument parameters

All parameters have been grouped in six categories:

- Expressivity: parameters that allow the physical player's expressivity to affect the instrument.
- Play Modes: parameters that act on the details of how the emulated player plays the instrument.
- Timbre: parameters that affect the instrument's timbre.
- Pitch: Master Tuning, Pitch, Transposition and microtonal settings.

- Advanced: additional parameters, especially for expert users.
- Room & Position / Effects: if the Ambiente Room Simulator parameter is set to ON, "Room & Position" opens Ambiente, the virtual room and positioning environment; otherwise, it shows the parameters affecting the audio signal processing applied at the final stage.

The most common and important parameters are presented on the main GUI.

Note: Currently, some parameters are "locked" on the iOS version. Unlocking through In-App Purchase will be provided in future.

Main controls

- Instrument Body (left/right arrows)
Use the arrows to select different instrument bodies with different timbres and characters.
- Bow Polyphony (if Play Mode = Bow)
[mapped to Key Switches B+C, B+C#, B+D, B+D#, B+E]
 - Mono String Crossing [B+C]: the instrument is monophonic; legato/portamento is achieved by overlapping two notes. If the legato/portamento involves two adjacent strings, the release of the previous note is stopped in order to avoid a "bichord" during the transition, depending on the interval set with the "Mono CrossString Muting" option (see Play Modes page).
 - Mono Poly Release [B+C#]: the instrument is monophonic; legato/portamento is achieved by overlapping two notes. If the legato/portamento involves two adjacent strings, the release of the first note is not stopped during the legato transition.
 - Double [B+D]: two simultaneous notes can be performed; legato and portamento is not possible if playing overlapping notes. The two selected notes are always on adjacent strings. Priority: last pressed note.
 - Double/Hold [B+D#]: two simultaneous notes can be performed; legato and portamento is possible. The two selected notes are always on adjacent strings. Priority: oldest note. The preferred pair of strings is selected using "Double Hold String Selection".
 - Auto [B+E]: two simultaneous initial notes can be performed; the first single note played turns the polyphony to Mono. Legato and portamento are also possible for two simultaneous notes. An additional latency is added at the attack in order to detect the double/mono mode. It is possible to change the

detection time acting on the "Auto Poly Detection Time" parameter in the Advanced section.

- Pizzicato Poly (if Play Mode = Pizzicato or Col Legno)
[mapped to Key Switches B+F, B+G]
 - Mono String Crossing: the instrument is monophonic; legato/portamento is achieved by overlapping two notes. If the legato/portamento involves two adjacent strings, the release of the previous note is stopped in order to avoid a "bichord" during the transition, depending on the interval set with the "Mono CrossString Muting" option (see Play Modes page).
 - Polyphony: it's possible to play multiple notes if they are distributed across different strings.
- Harmonics
[mapped to Key Switch F#]
 - OFF: no overtone.
 - 2: first overtone (one octave higher).
 - 3: second overtone (one octave and a fifth higher).
 - 4 Control: overtone selected by the "Harmonics 4 Control" parameter.
- Alternate Fingering
[mapped to Key Switch D#]
Selects the default position of the left hand on the fingerboard. Note: in order to obtain a portamento on the same string, it is required to check and eventually set the proper AltFing position of the starting note (BEFORE pressing the starting note) and of the ending note (BEFORE pressing the ending note).
 - Mid Position: left-hand stays in the middle of the neck.
 - Near the Bridge: left-hand stays towards the bridge (high position).
 - Near the Nut + Open: left-hand stays near the nut; it is possible to play open strings. Note: vibrato is not possible on open strings.
- Bow/Pizz Position: position of the bow or finger between "sul ponticello" and "sul tasto".
- Bow Pressure: (if Play Mode = Bow) "Weight" of the bow on the string.

- Pizz/C.Legno Tone: (if Play Mode = Pizzicato or Col Legno) Color of the pizzicato pluck or Col Legno strike.
- Tremolo
[mapped to Key Switch G#]
 - OFF: no tremolo.
 - Slow: slow tremolo.
 - Fast: fast tremolo (1.5 x slow tremolo rate).
Tremolo speed and behavior are controlled by the "Tremolo Min Speed" and "Tremolo Mode" parameters on the Play Modes / Right Hand section. A "manual" tremolo is possible using the Key Switch C# ("Manual Bowing: KeySwitch" parameter set to "Tremolo") or by holding Sustain ("Manual Bowing: Sustain+Note" set to "Tremolo").

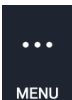
Note: even if Gesture Mode is set to Bipolar mode, when Tremolo is active the Gesture Mode is forced to Expression, because the bow change is defined by the auto-tremolo algorithm.
- Bow Lift
[mapped to Key Switch E]
Selects the starting/ending state of the bow. Note: the BowLift parameter acts mainly on the release, stopping the vibration of the string when it is set to "On String". It also acts slightly on the attack, but if you want to obtain a scratchy sound you should act on the velocity and on the bow pressure.
 - Off String: lift the bow away from the strings during release
 - On String: leaves the bow on the string during release
- Play Mode
[mapped to Key Switch C]
 - Bow: the instrument is played using the bow
 - Pizzicato: the instrument is played plucking the strings using the fingers
 - Col Legno: the instrument is played striking the strings with the bow
- Sordino
 - OFF: no mute applied
 - ON: a sordino is applied to obtain a muted sound

- RESET:
 - Initialize: reset the instrument and return to its initial state; this includes sound engine, MIDI mapping, micro tuning and room
 - Advanced:
 - Reload “current preset”: reload the current selected preset, losing all changes made from the latest saved settings
 - Sound Engine: reset or load sound engine parameters
 - MIDI Mapping: reset or load MIDI mapping settings
 - Micro Tuning: reset or load micro tuning settings
- PANIC: use this in case of stuck notes

Audio section

- Room & Position / Effects: if the Ambiente Room Simulator parameter is set to ON, “Room & Position” opens Ambiente, the virtual room and positioning environment; otherwise, it shows the parameters affecting the audio signal processing applied at the final stage.
- Pan Pot: pans the output signal left and right. Available only if Ambiente Room Simulator is OFF
- Main Volume: main output volume. This can be either 'Pre Room' or 'Post Room', depending on the 'Main Volume Processing' parameter located in the Advanced / ROOM page. Selecting 'Pre Room' means the main volume will only affect the source signal, leaving the room ambiance unaltered. Conversely, selecting 'Post Room' means the main volume will affect the overall output, including both the source signal and the room ambiance.

Header section

- SWAM logo : clicking or touching the SWAM logo opens the About section
- Main menu icon : opens the main menu page that provides access to:

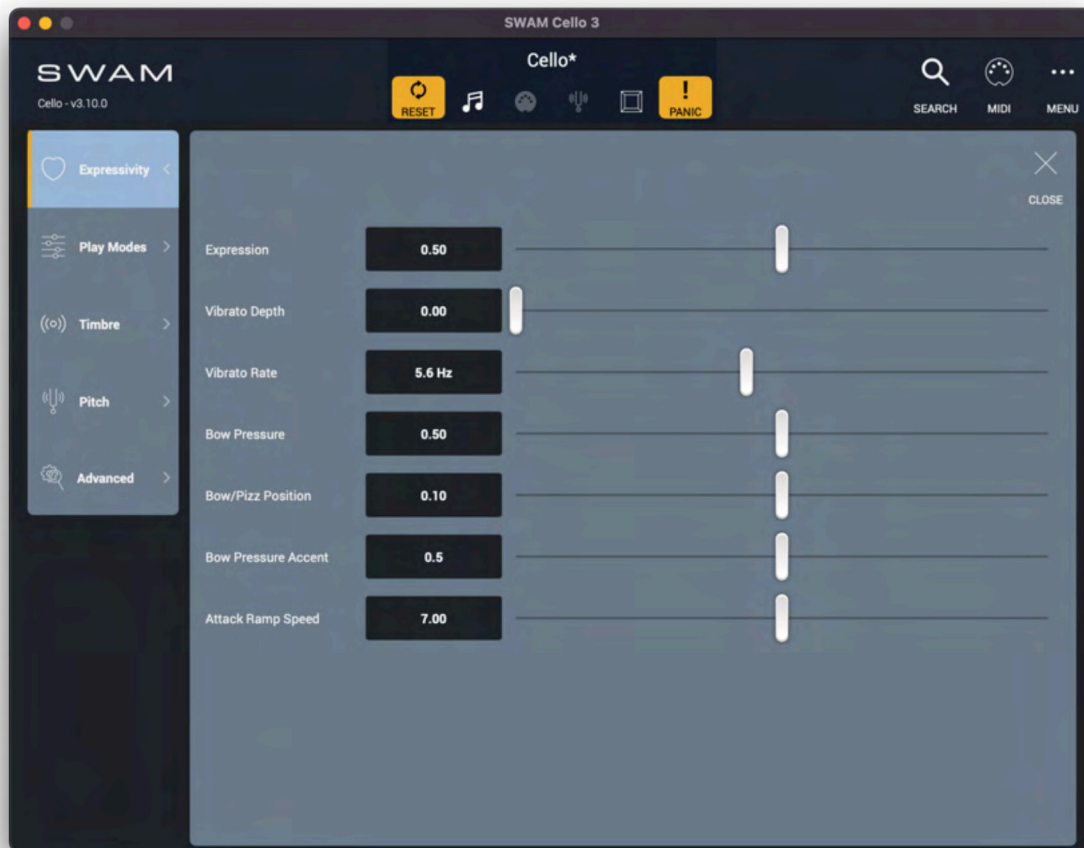
- Save & Load
- Reset
- Controller Mapping
- Parameter Lock
- Import
- Export
- Settings

- MIDI Mapping icon : opens MIDI mapping management section.

- SEARCH icon : opens the search screen for browsing parameters all at once with a big picture of the entire system.

- Playing Surface icon (only for iPadOS version) : open the playing surface for playing the instrument on the iPad glass.
NOTE: on iPhone, the playing surface is available rotating the device to landscape orientation.

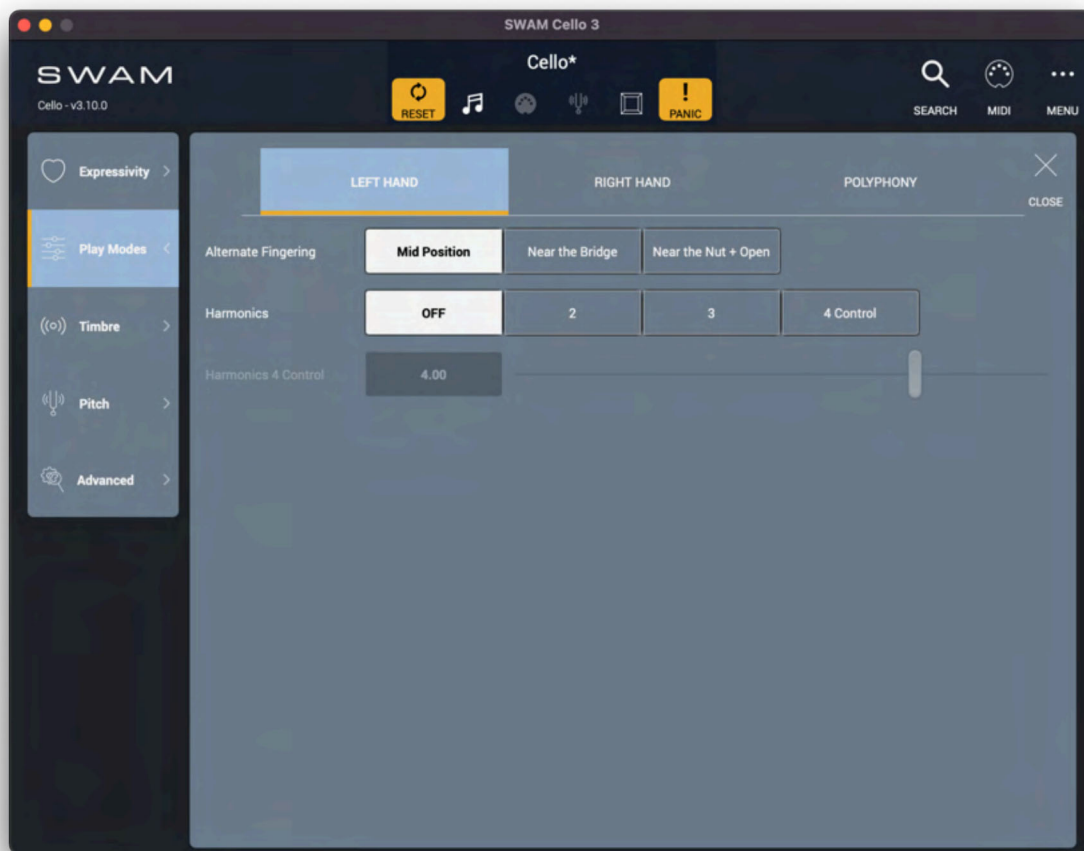
Expressivity



- Expression: main expressive control. On bowed strings, it's mostly related to the bow speed which in turn controls the dynamics.
- Vibrato Depth: controls the amount of vibrato.
- Vibrato Rate: controls the average frequency of the Vibrato applied to the model.
- Bow Pressure: "weight" of the bow on the string.
- Bow/Pizz Position: position of the bow or finger between "sul ponticello" and "sul tasto".
- Bow Pressure Accent: accent amount as a function of the velocity.
- Attack Ramp Speed (*locked on iOS*): steepness of the attacks, when controlled by Note-on Velocity.

Play Modes

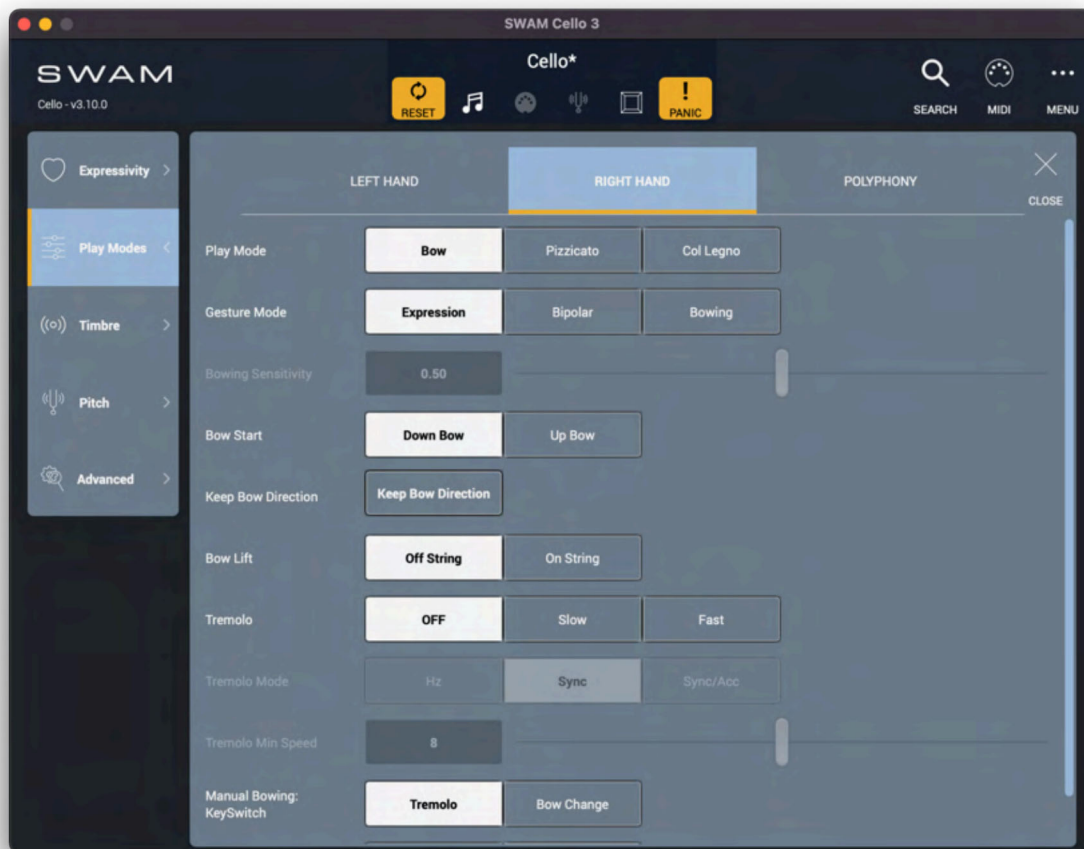
Left Hand



- Alternate Fingering
[mapped to Key Switch D#]
selects the default position of the left hand on the fingerboard. Note: in order to obtain a portamento on the same string, it is required to check and eventually set the proper AltFing position of the starting note (BEFORE pressing the starting note) and of the ending note (BEFORE pressing the ending note).
 - Mid: left-hand stays in the middle of the neck.
 - Bridge: left-hand stays towards the bridge (high position).
 - Nut+Open: left-hand stays near the nut (it is possible to play open strings). Vibrato is not possible on open strings.

- Harmonics
[mapped to Key Switch F#]
 - OFF: no overtone.
 - 2: first overtone (one octave higher).
 - 3: second overtone (one octave and a fifth higher).
 - 4 Control: overtone selected by the "Harmonics 4 Control" parameter.
- Harmonics 4 Control (*locked on iOS*)
Selects the overtone selected, when "Harmonics" is set to "4 Control"

Right Hand



- Play Mode:
[mapped to Key Switch C]

- Bow: the instrument is played using the bow
- Pizzicato: the instrument is played plucking the strings using the fingers
- Col Legno: the instrument is played striking the strings with the bow
- Gesture Mode: select the way the Expression is executed for PlayMode = Bow.
[mapped to Key Switch D]
 - Expression: expression is mainly interpreted as the speed of the bow in relation to the bow pressure. If the Expression MIDI CC Min Value is set to 0 (zero) in the MIDI Mapping Curve page, bow change automatically occurs when expression is set to zero for a while. Otherwise, if the Expression MIDI CC Min Value is set to a value different from zero in the MIDI Mapping Curve page, Auto-bow-change never occurs.
 - Bipolar: expression is interpreted as the speed of the bow in a bipolar way; zero speed coincides with CC value = 63. Values below 63 are interpreted as down-bow, while values over 63 are interpreted as up-bow.
In order to use this gesture, the MIDI mapping curve for Expression must be set to "Bipolar" as well. A warning message is displayed otherwise. See "Controller Mapping" section for more details.
Note: this gesture is not suitable for Breath and Wind Controllers. It is intended for pedals, wheels, ribbons, X-Y pads and similar.
 - Bowing: expression is interpreted as the displacement of the bow, the speed is proportional to the speed of the controller change. See also the "Bowing Sensitivity" parameter. Note: this gesture is not suitable for Breath and Wind Controllers. It is intended for pedals, wheels, ribbons, X-Y pads and similar.
- Bowing Sensitivity: when "Gesture Mode" is set to Bowing, this parameter controls how sensitive the bow speed is to the Expression changes.
- Bow Start:
[mapped to Key Switch F]
direction of the bow for staccato notes
 - Down Bow
 - Up Bow
- Keep Bow Direction:
[mapped to Key Switch G]
maintains the current bowing direction constant.

- Bow Lift:
[mapped to Key Switch E]
Selects the starting/ending state of the bow. Note: the “Bow Lift” parameter acts mainly on the release, stopping the vibration of the string when it is set to “On String”. It also acts slightly on the attack, but if you want to obtain a scratchy sound you should act on the velocity and on the bow pressure.
 - Off String: lift the bow away from the strings during release
 - On String: leaves the bow on the string during release
- Tremolo:
[mapped to Key Switch G#]
 - Off: no tremolo.
 - Slow: slow tremolo.
 - Fast: fast tremolo (1.5 x slow tremolo rate).

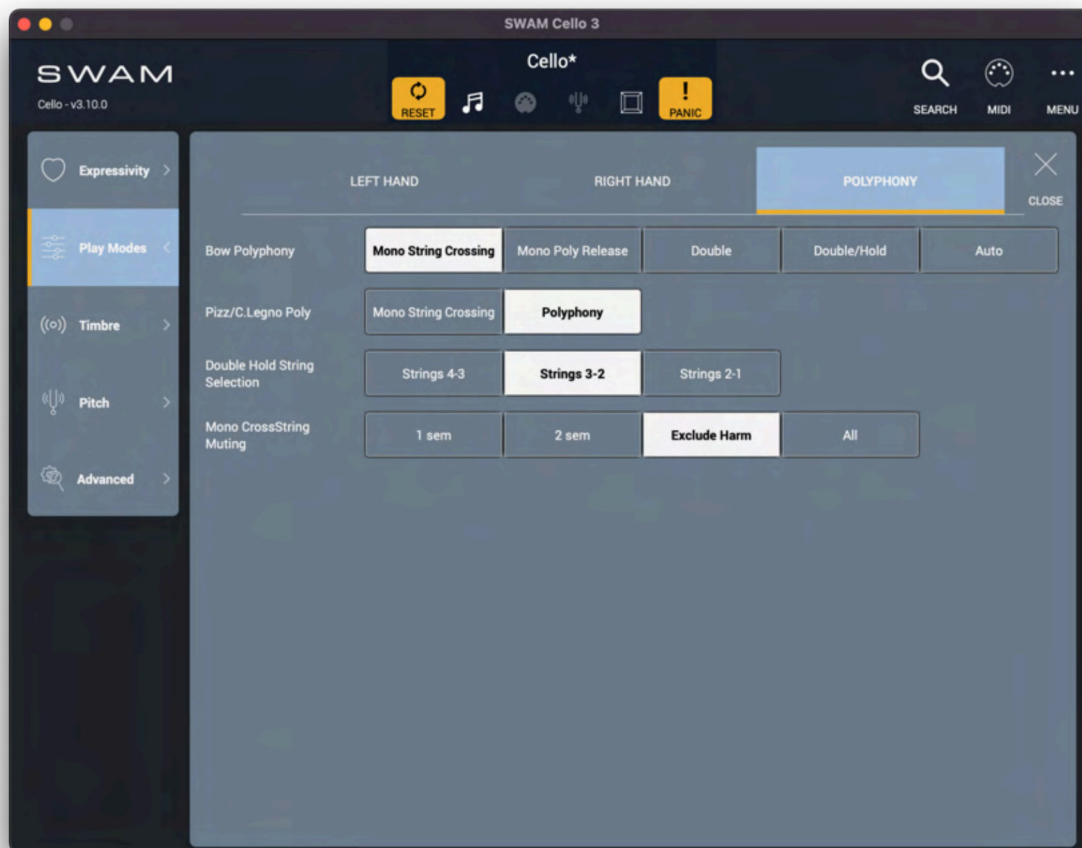
Tremolo speed and behavior are controlled by the “Tremolo Min Speed” and “Tremolo Mode” parameters on the Play Modes / Right Hand section. A “manual” tremolo is possible using the Key Switch C# (“Manual Bowing: KeySwitch” parameter set to “Tremolo”) or by holding Sustain (“Manual Bowing: Sustain+Note” set to “Tremolo”).

Note: even if Gesture Mode is set to Bipolar mode, when Tremolo is active the Gesture Mode is forced to Expression, because the bow change is defined by the auto-tremolo algorithm.

- Tremolo Mode: selects the tremolo rate behavior, affected by “Tremolo Min Speed” value:
[mapped to Key Switch A]
 - Hz: the tremolo rate is not synchronized with the current project BPM. Rate is exactly equal to “Tremolo Min Speed” for “Slow” tremolo, or 1.5 times for “Fast” tremolo.
 - Sync: the tremolo rate is synchronized with the current project BPM, using a rate value that is just above “Tremolo Min Speed”. Custom accents can be performed acting on the expression (expression “spikes” on the desired strokes).

- Sync/Acc: the tremolo rate is synchronized with the current project BPM, using a rate value that is just above “Tremolo Min Speed”; an accent is performed every four or three strokes, depending on the selected note length (plain vs triplets).
- Tremolo Min Speed: speed of the tremolo for the “Slow” option (see the Tremolo multi-state button and KeySwitch). If the “Tremolo Mode” is set to “Hz”, this value coincides with the “Slow” tremolo rate; the “Fast” tremolo rate is 1.5 times the “Slow” one. If the “Tremolo Mode” is set to “Sync” or “Sync/Acc”, this value is a threshold: the tremolo is synchronized with current project BPM selecting a rate just greater than the threshold. The actual value of the tremolo rate is expressed as note length (e.g. 1/8 means quavers, 1/8T means quaver triplets, 1/16 means semiquavers, 1/16T means semiquaver triplets, and so on). The “Fast” rate is just the shorter note length just below the “Slow” selected value (e.g. if “Slow” corresponds to 1/8T, “Fast” is 1/16).
- Manual Bowing KeySwitch: sets the behavior for the C# Key Switch:
 - Tremolo: C# K.S. is set to “Manual Tremolo”; a bow change is performed on both Note-on and Note-off; the velocity of the Key Switch is ignored.
 - Bow Change: C# K.S. is set to “Bow Change”, a bow change is performed just on Note-on; the velocity of the Key Switch influences the accent of the bow stroke.
- Manual Bowing Sustain+Note: sets the behavior of the Note-on / Note-off of the currently played note, when Sustain is active:
 - Tremolo: a bow change is performed on both Note-on and Note-off
 - Bow Change: a bow change is performed just on Note-on.

Polyphony



- Bow Polyphony (if Play Mode = Bow)
[mapped to Key Switches B+C, B+C#, B+D, B+D#, B+E]
 - Mono String Crossing [B+C]: the instrument is monophonic; legato/portamento is achieved by overlapping two notes. If the legato/portamento involves two adjacent strings, the release of the previous note is stopped in order to avoid a “bichord” during the transition, depending on the interval set with the “Mono CrossString Muting” option (see Play Modes page).
 - Mono Poly Release [B+C#]: the instrument is monophonic; legato/portamento is achieved by overlapping two notes. If the legato/portamento involves two adjacent strings, the release of the first note is not stopped during the legato transition.

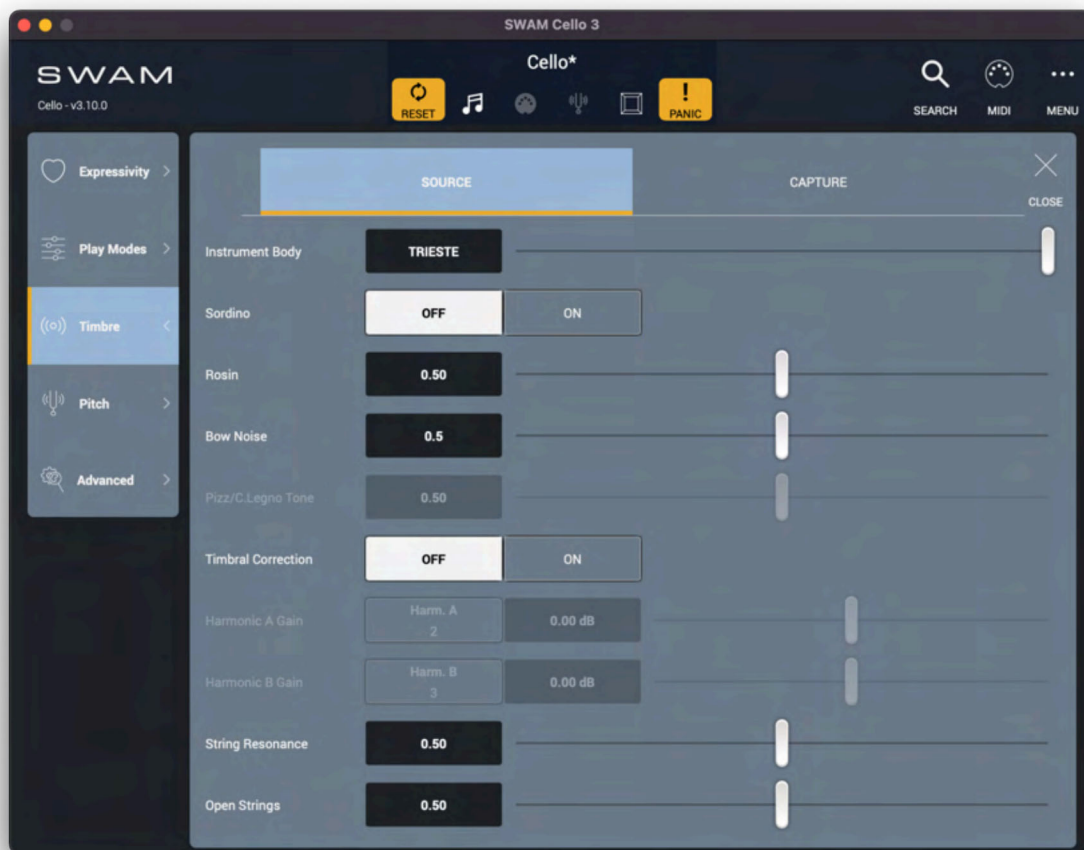
- Double [B+D]: two simultaneous notes can be performed; legato and portamento is not possible if playing overlapping notes. The two selected notes are always on adjacent strings. Priority: last pressed note.
- Double/Hold [B+D#]: two simultaneous notes can be performed; legato and portamento is possible. The two selected notes are always on adjacent strings. Priority: oldest note. The preferred pair of strings is selected using "Double Hold String Selection".
- Auto [B+E]: two simultaneous initial notes can be performed; the first single note played turns the polyphony to Mono. Legato and portamento are also possible for two simultaneous notes. An additional latency is added at the attack in order to detect the double/mono mode. It is possible to change the detection time acting on the "Auto Poly Detection Time" parameter in the Advanced section.
- Pizz/C.Legno Poly (if Play Mode = Pizzicato or Col Legno)
[mapped to Key Switches B+F, B+G]
 - Mono String Crossing: the instrument is monophonic; legato/portamento is achieved by overlapping two notes. If the legato/portamento involves two adjacent strings, the release of the previous note is stopped in order to avoid a "bichord" during the transition, depending on the interval set with the "Mono CrossString Muting" option (see Play Modes page).
 - Polyphony: it's possible to play multiple notes if they are distributed across different strings.
- Double Hold String Selection: sets the preferred pair of strings for double stops
[mapped to Key Switches B+F#, B+G#, B+A#]
 - Strings 4-3: lowest strings (Key Switches B+F#)
 - Strings 3-2: middle strings (Key Switches B+G#)
 - Strings 2-1: higher strings (Key Switches B+A#)
- Mono CrossString Muting: affects the transitions across adjacent strings when the "Bow Polyphony" parameter is set to "Mono String Crossing". The release of the "old" string gets stopped if the interval performed between the first and the second note is included in the interval specified by the option:
 - 1 sem: 1 semitone (minor 2nd) for the first option
 - 2 sem: 1 or 2 semitones (minor 2nd and major 2nd) for the second option

- Exclude Harm: 1, 2, 3, 4, 6, 8, 9, 11, 13 semitones (minor and major 2nd, minor and major 3rd, tritone, minor and major 6th, major 7th, minor 9th) for the third option
- All: always for the fourth option

For all the remaining intervals the “old” string is left free to vibrate, creating two overlapping tones during the transition.

Timbre

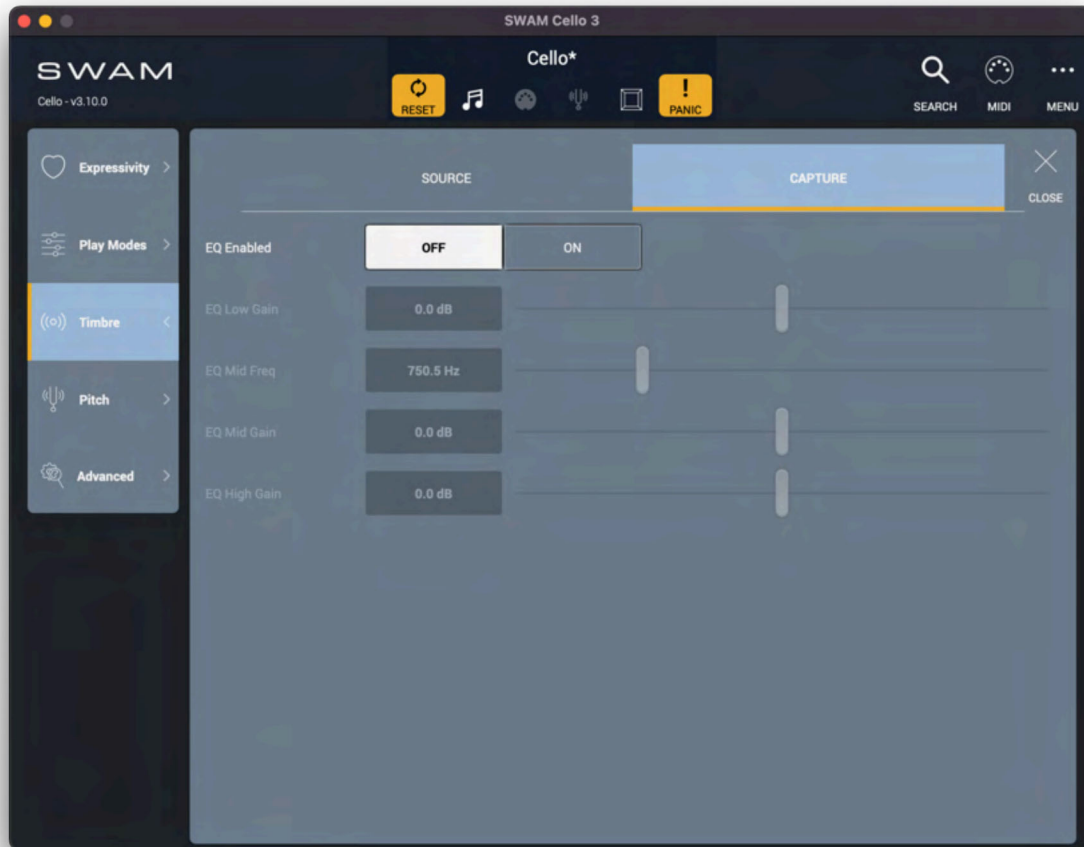
Source



- Instrument Body: selects different instrument bodies with different timbres and characters.
- Sordino:

- OFF: no mute applied
- ON: a sordino is applied to obtain a muted sound
- Rosin (*locked on iOS*): amount of rosin, affects the stickiness between the bow and the string.
- Bow Noise: amount of noise produced by the bow rubbing the string.
- Pizz/C.Legno Tone: color of the pizzicato pluck or Col Legno strike (enabled when *Play Mode* is set to *Pizzicato* or *Col Legno*).
- Timbral Correction: enables or disables the Harmonics Timbral Shaping.
 - Harmonic A Gain: the gain amount for harmonic selected by "Harm. A"
 - Harmonic B Gain (*locked on iOS*): the gain amount for harmonic selected by "Harm. B"
- String Resonance: resonance amount of the string. It affects the release time, the open strings vibration and, partially, the timbre.
- Open Strings: amount of sympathetic effect (resonance) produced by open strings.

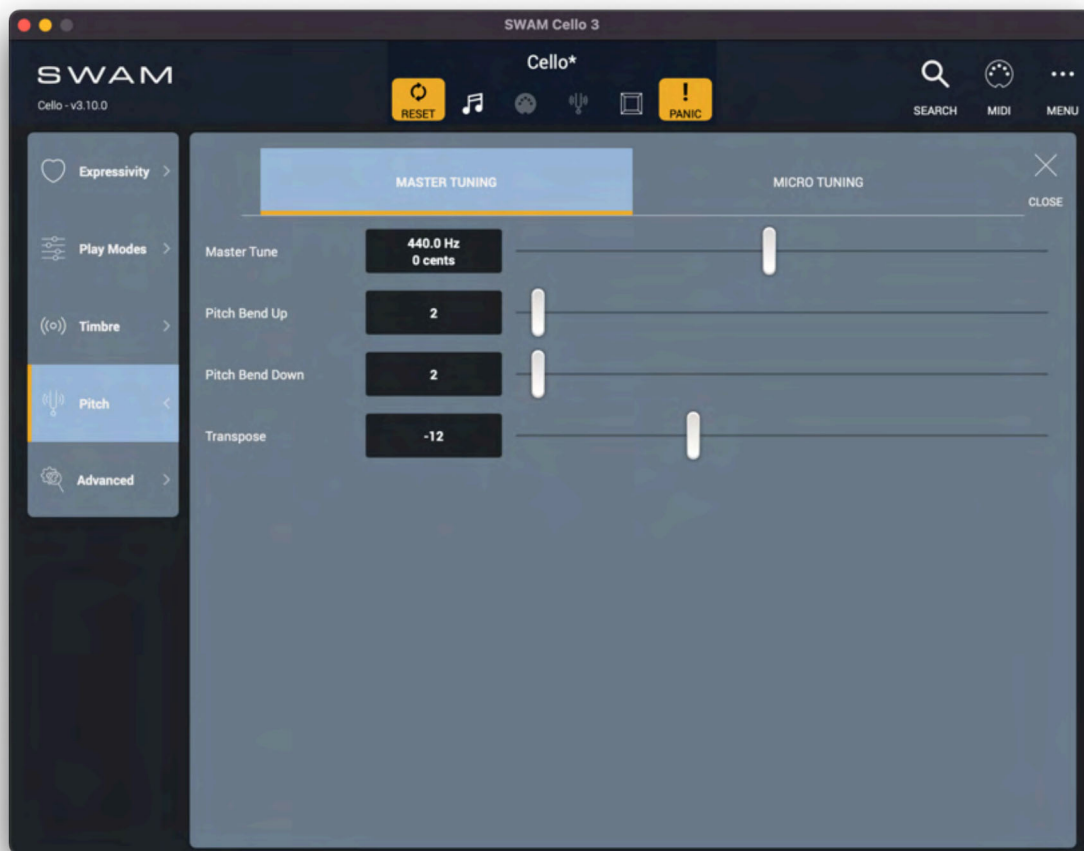
Capture



- EQ Enabled: enables or disables the Equalizer (see next).
- EQ Low Gain, EQ Mid Gain, EQ Mid Freq, EQ High Gain: Equalizer parameters. Note that ranges are different for each instrument. They have been calibrated for the harmonic contents of each instrument.

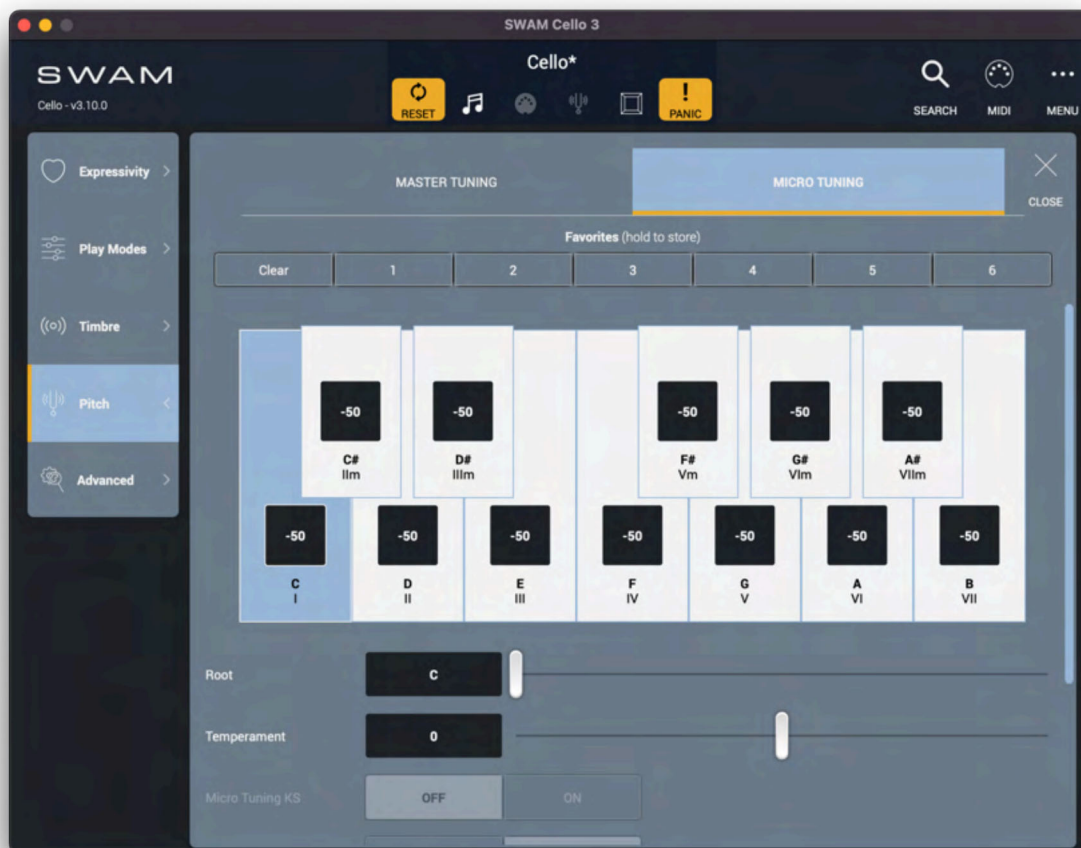
Pitch

Master Tuning



- Master Tune: master tuning of the instrument displayed both in *Hz* and *cents*.
- Pitch Bend Up: the maximum amount of upwards pitch bend, in *semitones*.
- Pitch Bend Down: the maximum amount of downwards pitch-bend, in *semitones*.
- Transpose: the amount to transpose the instrument, in *semitones*

Micro Tuning



- Favorites: it is possible to store up to 7 different settings. Click and hold the preset slot to store the current configuration; click on any preset slot to recall the preset. Click on the Clear button to reset the current configuration to the default state
- Micro Tuning Keys:
 - Activation: Click on any key to toggle the detune, by the amount shown on the button.
 - Edit: Click on the detune value to adjust. The value is presented in cents (hundredths of a semitone)
- Root: select the root note of Micro Tuning scale

- Temperament: if set higher than zero, this applies a kind of “spread intonation”:
 - it does not affect the pitch in the middle of the instrument
 - the higher the pitch, the more upward detuning is applied
 - the lower the pitch, the more downward detuning is applied

This means that the overall intonation of the instrument can be “spaced apart”.

If set lower than zero, it does the opposite: it applies a kind of “compressed intonation”

- Micro Tuning KS:
 - OFF: disable key switches for micro tuning
 - ON: enable key switches for micro tuning
- Reset Micro Tuning KS:
 - OFF Micro Tuning KS activation does not reset or affect current detuning
 - ON: Micro Tuning KS activation resets the detuning applied.
- Cavit Scale Ultimate (by Cavit Artanlar): enables the “SWAM Micro Tuning” SysEx protocol, which allows “Cavit Scale Ultimate” iOS app to control micro tuning and transposition, and to apply musical nuances typical of Arabic and Eastern music.
- MAQAM (by Mazeka Toys - mazekatoys.com):
 - OFF: remove the factory MIDI MAQAM configuration
 - ON: apply MAQAM factory mapping (note: all mappings to parameters assigned to CC from 102 to 114 will be removed). This allows the use of a MAQAM remote controller to apply Micro Tuning and Transposition on the fly.
Note: be sure MAQAM is connected properly through a MIDI interface so that MIDI events can reach the SWAM application or plugin.
- MAQAM Transpose:
 - Note: control keyboard transpose
 - Root: control micro tuning transpose



Applying micro tuning through SysEx

It's possible to control Micro Tuning and Transposition through MIDI SysEx messages. SWAM responds to a few SysEx protocols: [SWAM Micro Tuning protocol](#), [Custom protocol](#) and [Yamaha XG arranger protocol](#).

SWAM Micro Tuning protocol

Note: Be sure "Cavit Scale Ultimate" switch is set to ON in the Pitch / Micro Tuning view.

Byte 1 - Beginning of sysex: F0H
Bytes 2, 3, 4 - Audio Modeling Manufacturer ID: 00H 21H 2DH
Bytes 5, 6 - Model ID: 00H 00H
Byte 7 - Command type: 00H (Read), 01H (Write)
Byte 8 - Category: 00H (System)
Byte 9 - Parameter: 00H (Transpose), 01H (Micro Tuning)
... - Data bytes
Byte N - End of sysex message: F7H

Data bytes

Transpose (1 byte):

Byte 10 - Transposition value in semitones: 00H -> 7FH
 < 40H: negative transposition
 40H: 0 cents
 > 40H positive transposition

Micro Tuning (3 bytes):

Byte 10 - MIDI note number: 00H -> 7FH
Byte 11 - Sign:
 <= 40H positive
 > 40H negative
Byte 12 - Micro Tuning value in cents: 00H -> 64H (0 to 100 cents)



Custom Protocol

This protocol allows to change both Micro Tuning and Transposition on the fly through SysEx messages.

1) Micro Tuning message

Each message is 4 bytes:

1st byte: Beginning of sysex (F0)
2nd byte: Note value (00=C, 02=C#, 04=D, ..., 12=A, 14=A#, 16=B)
3rd byte: Detuning value (00 = 0, 4E = -50)
4th byte: end of sysex (F7)

F0 10 10 F7: reset all notes to not detuned

Examples:

F0 00 4E F7: Note C = -50 cents
F0 10 4E F7: Note G# = -50 cents
F0 16 00 F7: Note B = 0 cents

2) Transpose message

Each message is 3 bytes:

1st byte = Beginning of sysex (F0)
2nd byte = Transpose value (37=-9, 38=-8, 39=-7, 3A=-6 ... 40=0 ... 48=+8, 49=+9)
3rd byte = End of sysex (F7)

F0 10 F7: reset transpose to 0

Yamaha XG arranger protocol

Each message is 9 bytes:

1st byte: Beginning of sysex (F0)
2nd byte: Manufacturer ID - Yamaha (43)
3rd byte: Device ID (values between 0x10 and 0x1F are allowed)
4th byte: Model ID - XG (4C)
5th byte: Address High (08)
6th byte: Address Mid (00)
7th byte: Address Low - Note value (41=C, 42=C#, 43=D, ..., 4C = B)
8th byte: Detuning value (04=-60, 05=-59, 06=-58, ..., 40=0, ..., 7B=+59, 7C=+60)
9th byte: End of sysex (F7)

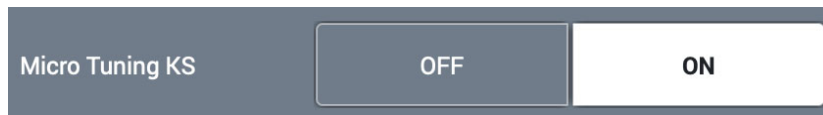
Examples:

F0 43 10 4C 08 00 41 5E F7: Note C = +30 cents
F0 43 10 4C 08 00 41 18 F7: Note C = -40 cents
F0 43 10 4C 08 00 4A 0E F7: Note A = -50 cents

Applying micro tuning using key switches

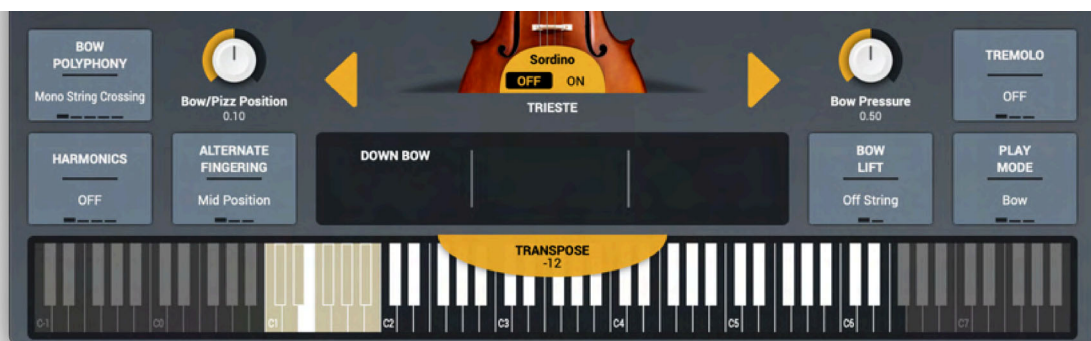
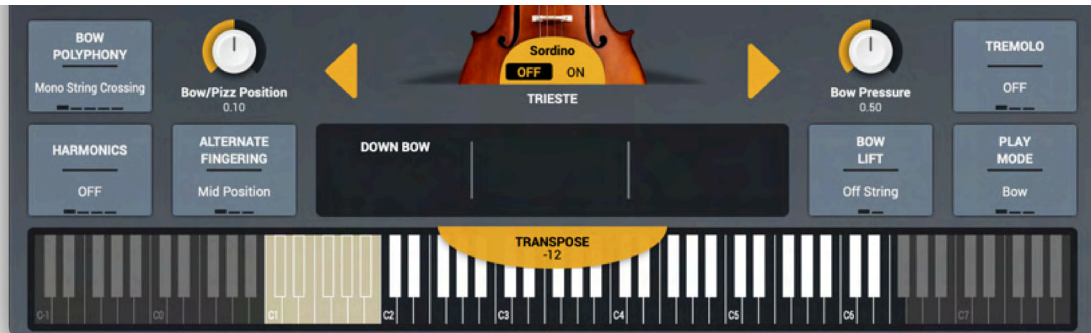
You can also apply micro tuning to notes using key switches. To do this you must first go to Advanced -> MIDI and enable the key switches using the "KS Octave" parameter.

Then go to Pitch -> MICRO TUNING and set the "Micro Tuning KS" parameter to "ON".



This parameter can be assigned to a MIDI Control Change. A handy mapping is CC "64", i.e. the Sustain pedal, which is the most suitable option for this task.

When "Micro Tuning KS" is active, you can press the appropriate key switches that correspond to the notes you would like to detune. For example, by pressing the E1 key switch detuning will be applied to all E notes.



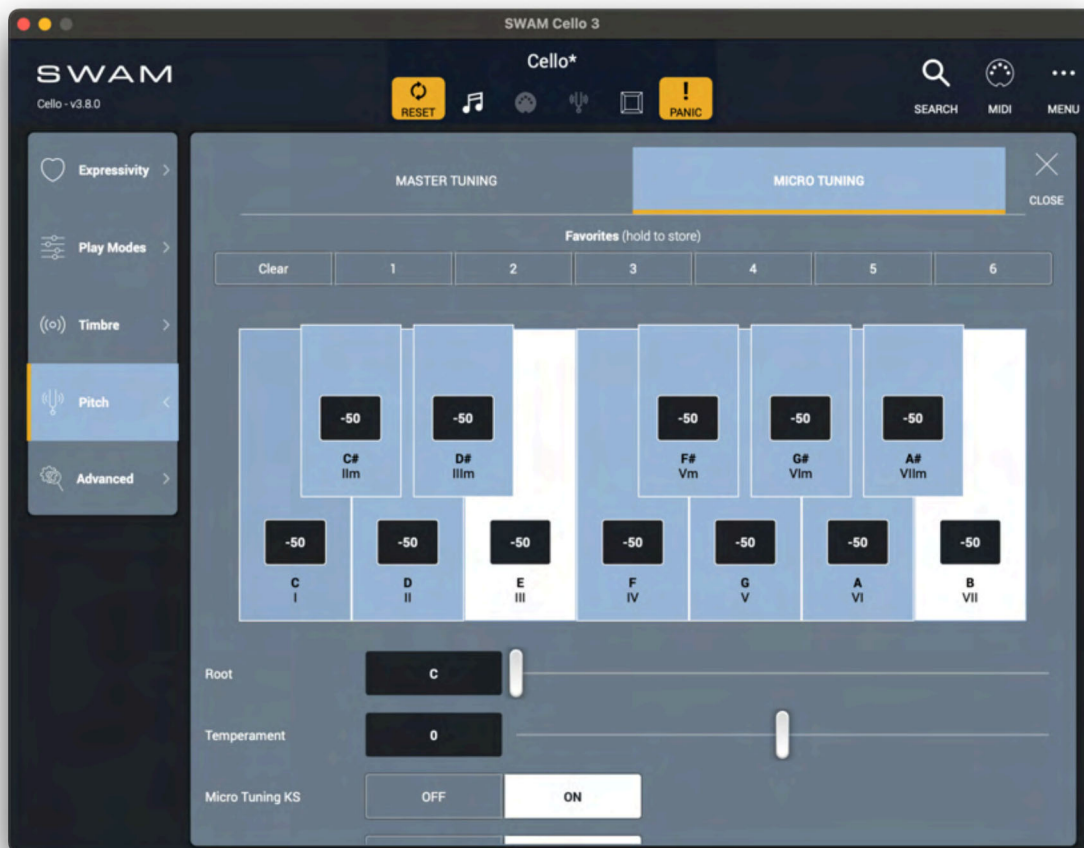
How to detune selected notes

1. Set the "Micro Tuning KS" parameter ON by pressing and holding down the Sustain pedal (or send a value greater than 64 via the CC you have assigned for Micro Tuning KS); the key switches will turn light gray.
2. Press any key switches (even multiple at the once) that correspond to the notes you want to detune.
3. Exit "setup" mode by releasing the sustain pedal (or send a value below 64 via the CC assigned for this task).

For example: if you need an Arabic scale in C and want to detune all E and B notes -50 cents, the steps would be:

1. Press and hold the sustain pedal
2. Press E1 and B1
3. Release the sustain pedal

The affected notes and the detune amount is displayed in the Pitch -> MICRO TUNING page.



If the "Reset Micro Tuning KS" parameter is ON, as soon as you re-enable Micro Tuning KS (by pressing the Sustain pedal or sending a value greater than 64 via the CC you have assigned for this task), the current detuning of all the notes will be cleared.

Keeping the pedal pressed, you can immediately detune a new set of notes by pressing the appropriate key switches.

This allows you to apply different micro tuning scales on the fly.

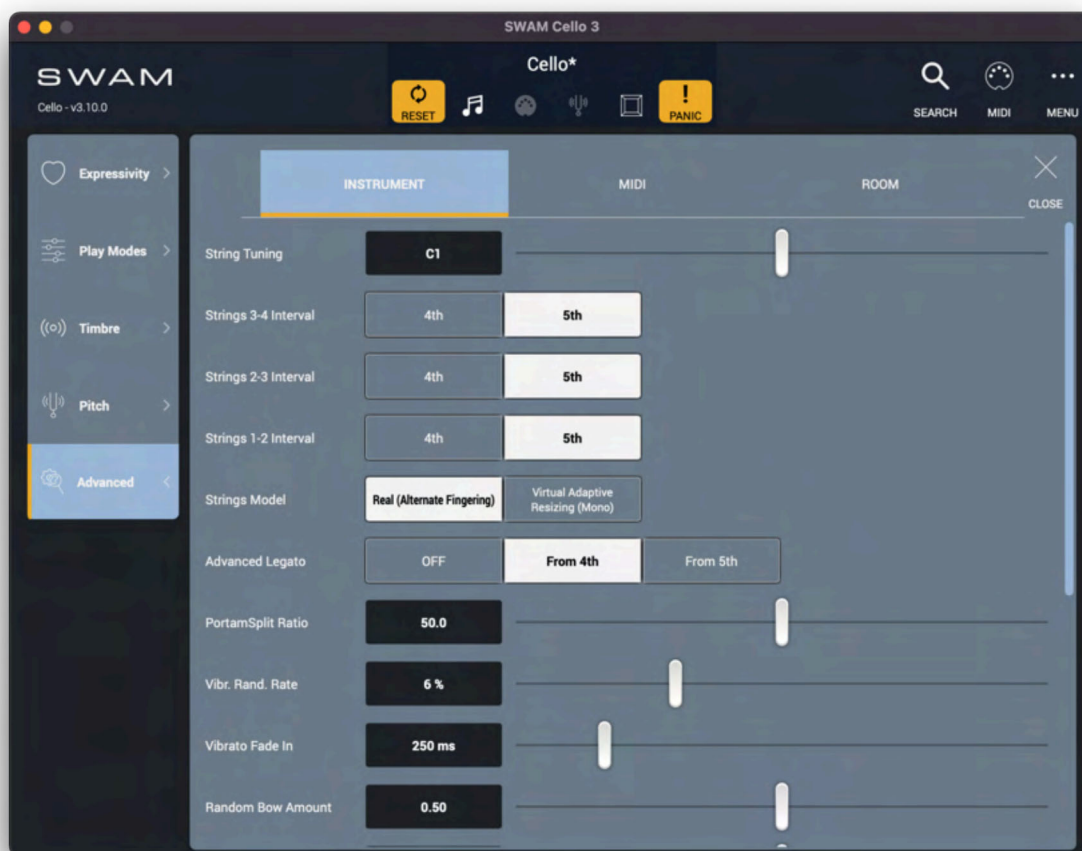
If you don't want to reset the micro tuning every time you re-enable Micro Tuning KS set "Reset Micro Tuning KS" parameter to OFF.

Micro Tuning Templates

From the Main "... Menu and Preset Manager it is possible to Save and Recall Micro Tuning Templates. A list of Factory Micro Tuning Templates is provided.

Advanced

INSTRUMENT



- String Tuning, Strings 3-4 Interval (locked on iOS), Strings 2-3 Interval (locked on iOS), Strings 1-2 Interval (locked on iOS): tuning of the lowest string followed by the tuning of the other strings. As in the real instrument, the tuning of the other strings can be set only a fourth or a fifth higher than the adjacent lower string.
- String Model
 - Real (Alternate Fingering): the model emulates a real instrument with four strings with fixed thickness and tension.
 - Virtual Adaptive Resizing (Mono): the model emulates an instrument with just a single string that ideally has different thickness and tension, depending on the actual played note. This way is possible to bend up from the lowest note to the highest one with no jumps: set Pitch Bend range to 48 semitones, play

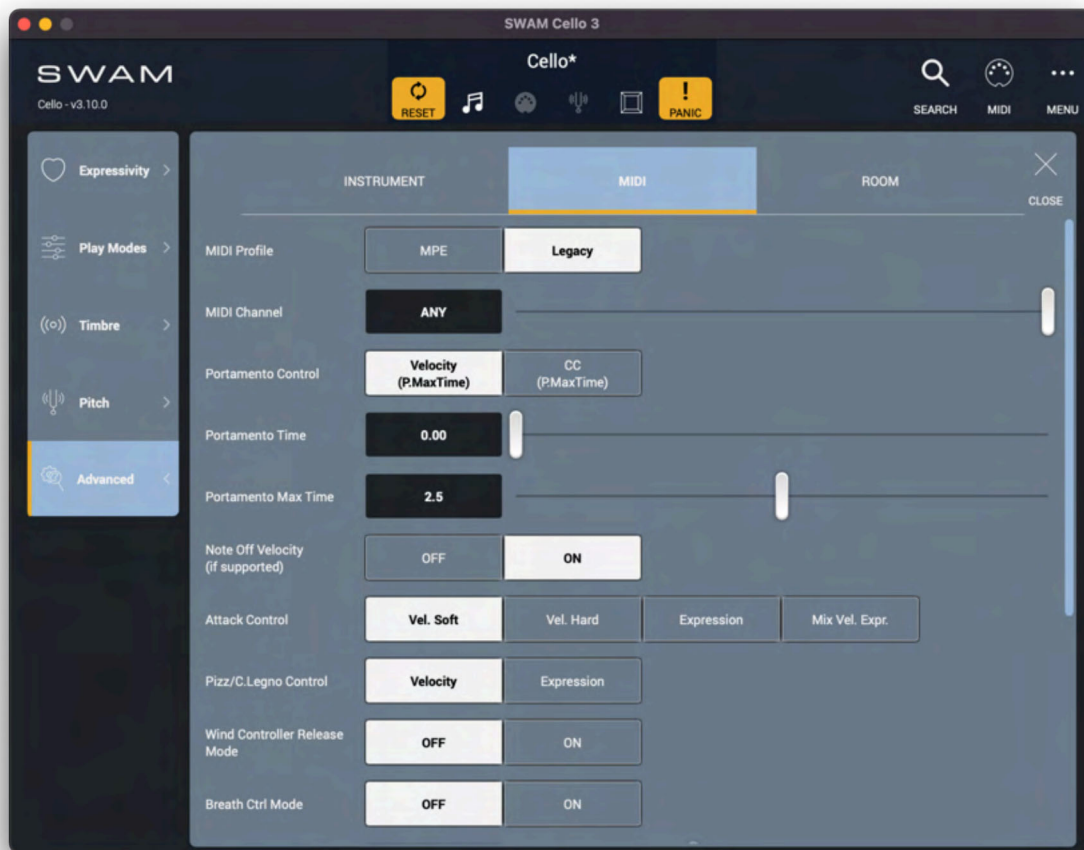
the lowest note, and move the Pitch Bend all the way up.

Since the instrument has just one string, polyphony is not possible (all parameters related to "Polyphony" are disabled).

- Advanced Legato (*locked on iOS*)
Legato is modeled with a sophisticated model that mimic the real behavior of
 - OFF: legato is modeled with a fast sweep transition.
 - From 4th: advanced legato starts from perfect 4th interval (5 semitones)
 - From 5th: advanced legato starts from perfect 5th interval (7 semitones)
- PortamSplit Ratio (*locked on iOS*): sets the split point (percentage of the interval) for a portamento across two strings. For example, 20 means 20% on the starting string and 80% on the other string.
- Vibrato Rate Rand (*locked on iOS*): controls how much randomness is applied to the rate of vibrato.
- Vibrato Fade In (*locked on iOS*): how long (in milliseconds) it takes for the vibrato to reach the final amount after each Note On.
- Random Bow Amount (*locked on iOS*): amount of randomness of the bow pressure, speed and position.
- Random Finger (*locked on iOS*): amount of randomness of the left-hand finger position (pitch).
- Dynamic Transitions (*locked on iOS*): controls the shape of the expression during legato transitions.
- Panpot Type (*locked on iOS*)
 - Dyn1, Dyn2 - emulates small movements of the player.
 - Acoustic - adjusts the position of the instrument in the sound field, but does not affect the reverb.
 - Balance - affects both the instrument and the reverb.
- Interactive Bow Compensation (*locked on iOS*): controls the amount of correction of the bow pressure to maintain a good vibrating sound.
- Pizzicato Fingerboard Interference (*locked on iOS*): controls the amount of interference between the string and the fingerboard during pizzicato.

- Auto Poly Detection Time: sets the look-ahead time for the detection of bichords when polyphony is set to "Auto". The detection of bichords only happens for staccato notes. Note: an extra-latency is added to staccato notes when polyphony is set to "Auto", so it is better to set it to the smallest possible value.
- Staccato Interval Time: minimum time separation between a note-off event and a consequent note-on event to be interpreted as "staccato" articulation. 18 ms is suitable for fast staccato notes, but higher execution precision is required.

MIDI



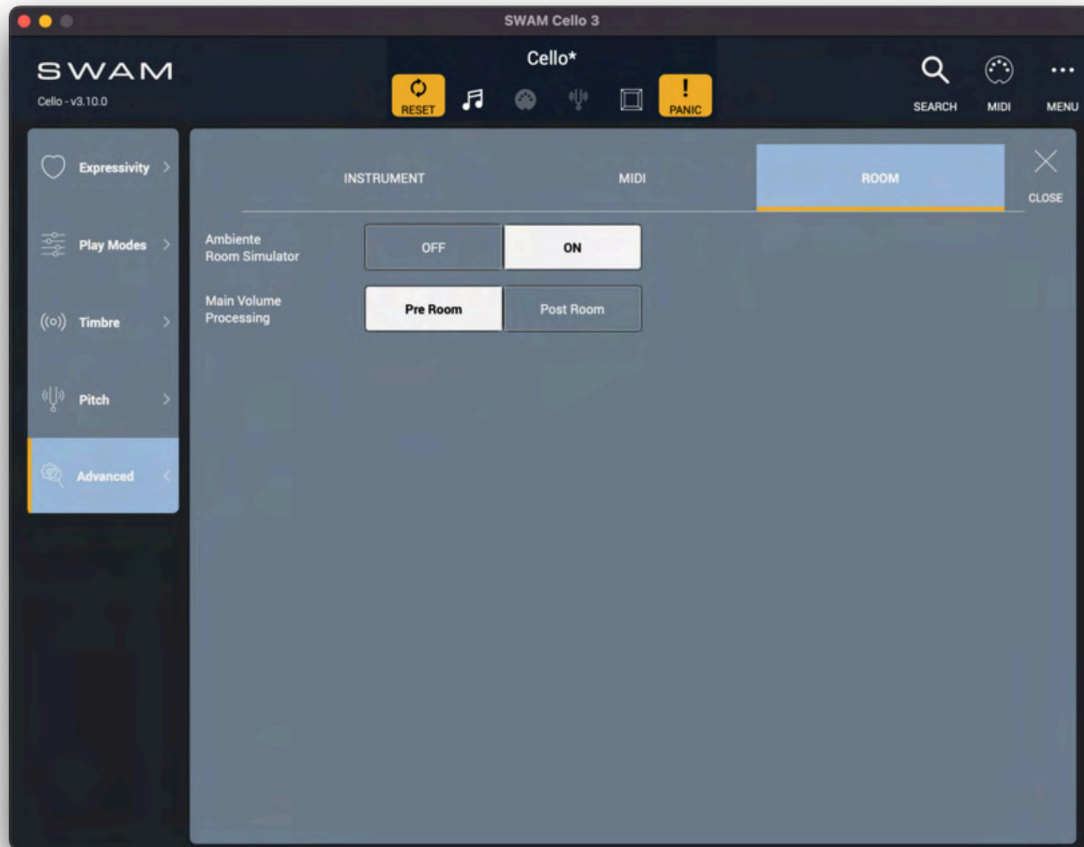
- MIDI Profile:
 - MPE: for MIDI Polyphonic Expression compatible devices.
 - Legacy: MIDI 1.0 devices.
- MIDI Channel: the MIDI channel the sound generator receives notes and controllers that affects the sound (default is ANY). Note that this can be different from Key

Switches MIDI Channel, so it is possible to control Key Switches from a different device on a different Channel. For MPE Devices this is forced to ANY.

- Portamento Control:
 - Velocity: the portamento time is controlled by the Note On velocity of overlapping notes.
 - CC: the portamento time is controlled by a MIDI Continuous Control (such as Control Change, AfterTouch, NRPN)
- Portamento Time: midi CC control for the portamento time, in relation to the max portamento time.
- Portamento Max Time: controls the maximum portamento time. Set it to OFF to disable portamento (no portamento when this value is < 1.1).
- Note Off Velocity (if supported):
 - OFF: fixed Note Off velocity at 112
 - ON: get legato retrigger velocity from note off velocity
- Attack Control - how the attack of the note is controlled:
 - Vel. Soft - controlled weakly by the Note On velocity.
 - Vel. Hard - controlled strongly by the Note On velocity.
 - Expression - controlled by the shape of expression control.
 - Mix Vel. Expr. - controlled by both the velocity and the initial expression.
See also "Attack Ramp Speed" for controlling the character of the attack.
- Pizz/C.Legno Control:
 - Velocity: control pizzicato or col legno play mode through velocity.
 - Expression: control pizzicato or col legno play mode through expression.
- Wind Controller Release Mode: useful especially for handling Wind Controllers, as they send Note Off events when Expression reaches the minimum value.
- Breath Ctrl Mode (Breath Control Mode):

- OFF: disabled.
- ON: suitable for Breath Controllers. When Play Mode is set to Bow, allows the player to produce another note attack, using the breath envelope, while holding a note.
- Breath Ctrl Hi-Res Threshold: (Breath Control Hi-Resolution Threshold) set the threshold for the Note On to be triggered when using MIDI Hi-Resolution messages for the Expression parameter, when Breath Control Mode is ON.
- Breath Ctrl Attack Sens.: (Breath Control Attack Sensitivity) controls the sensitivity of the Note On attack strength for notes triggered in Breath Control Mode.
- Expression String Release Muting:
 - ON: the release can be stopped setting the Expression to zero
 - OFF: for Wind Controllers, Breath Controllers and expressive controllers with expression release concurrent to the Note Off: set this parameter to "OFF", to avoid a sudden termination of the vibrating string release.
- KS MIDI Channel (Key Switches MIDI Channel): select which MIDI channel will be used to receive key switch information. Select from channel 1 to 16 or ANY.
- KS Octave (Key Switches Octave Transpose): allows you to transpose the Key Switches to begin in the octave from C-1 to C2, or turn off key switches (OFF).
- KS Velocity Remap (Key Switches Velocity Remapping): since some Key Switches are velocity-dependent, this parameter controls the distribution of the thresholds between states.

ROOM



- Ambiente Room Simulator:
 - OFF: Disables Ambiente. This option is recommended if you already have projects using that reverb or a third party external reverb.
 - ON: Activates Ambiente, the onboard virtual room and position engine. This option is recommended to get the most realistic sound for each instrument that is located in the same room.
- Main Volume Processing:
 - Pre Room: the volume control is applied to the sound source, but not to the virtual room. This ensures that reverb is not affected by eventual volume automation.
 - Post Room: the volume control is applied to the sound signal after the virtual room processing. In case of volume control, reverb is also affected.